

AI-Powered Multi-Vehicle Route Optimization for Context-Aware Deliveries

In today's rapidly evolving logistics ecosystem, minimizing delivery time while maximizing operational efficiency is increasingly critical. Factors such as customer unavailability, unpredictable weather, suboptimal route planning, and rigid delivery schedules often result in avoidable fuel consumption, excessive idle time, and diminished customer satisfaction. Conventional routing systems tend to use static inputs—like distance and average speed—without adapting to real-time conditions.

Our proposed solution is an intelligent **Multi-Vehicle Route Optimization Platform** that assigns delivery tasks based on customers' preferred time slots, captured prior to dispatch. Going beyond static heuristics, our system leverages **Google OR-Tools** enhanced by a machine learning layer that dynamically adjusts the cost matrix using contextual data such as:

- Real-time and historical traffic patterns
- Weather conditions (temperature, humidity, wind speed)
- Time of day (e.g., rush hours, night time)
- Package-specific attributes (e.g., weight)

Using APIs like **OpenWeather** and curated historical logistics data, the ML model predicts more accurate travel times, allowing for optimized and context-aware route planning.

Our implementation includes:

- A realistic simulation of the delivery environment using **OpenStreetMap**
- An **Admin Dashboard** for managing and assigning optimized delivery routes
- A **Delivery Agent Portal** that enables real-time task tracking and customer communication

By integrating predictive analytics into routing logic, our solution significantly reduces reliance on expensive external APIs, lowers operational costs, and enhances delivery reliability—delivering a more **efficient, sustainable, and customer-first logistics experience**.

<https://developers.google.com/optimization/routing/vrptw>

<https://osmnx.readthedocs.io/en/stable/>

RL MODEL

observation_space → a vector containing current location, time, undelivered package info

action_space → one discrete action per delivery (e.g., choose delivery 0 to $n-1$)

On-time delivery: +200

Late delivery: penalty based on how late

Distance penalty: $-5*$

Invalid action (already delivered): -20

More About Project

Main Purpose: Optimizes delivery routes for multiple delivery personnel, considering real-world road networks, time windows, and travel constraints using Google's OR-Tools Vehicle Routing Problem with Time Windows (VRPTW) solver.

Key Components:

1. **Real Road Network Integration:** Uses OSMnx to fetch actual road data for Udupi from OpenStreetMap
2. **Multi-Vehicle Routing:** Assigns deliveries to multiple delivery persons optimally
3. **Time Window Constraints:** Respects delivery time windows (when customers are available)
4. **Real Travel Time Calculation:** Uses NetworkX shortest path algorithms on actual road networks
5. **Interactive Visualization:** Creates detailed maps showing optimized routes with Folium

Workflow:

1. Loads/creates Udupi road network graph
2. Takes delivery persons' locations and delivery requests with time windows
3. Calculates travel time matrix between all locations using real roads
4. Uses OR-Tools to solve the optimization problem
5. Returns optimized routes with detailed turn-by-turn directions
6. Visualizes results on an interactive map

Integrating ML for Enhanced Time Prediction

Q) why ML?

Ans: OSMnx uses OpenStreetMap road types to assign speeds:
Hence the time is not entirely reliable.

ML-Enhanced Travel Time Predictor

Implementation Strategy

1. Data Collection Phase

- **Historical Route Data:** Collect actual delivery times vs predicted times
- **Environmental Data:** Weather, traffic conditions, time of day
- **Road Characteristics:** Road quality, intersection density, traffic lights

2. Feature Engineering

The ML model considers multiple factors:

- **Temporal:** Hour, day of week, rush hour, seasonal patterns
- **Environmental:** Weather conditions, temperature, humidity
- **Infrastructure:** Road types, intersection count, road quality
- **Traffic:** Historical traffic patterns, real-time density
- **Geographic:** Distance, route complexity

3. Model Integration Steps

Phase 1: Data Collection

Phase 2: Model Training

Phase 3: Integration

4. External Data Sources to Integrate

- **Weather API:** OpenWeatherMap, AccuWeather
- **Traffic API:** Google Maps Traffic API, HERE Traffic API
- **Historical Data:** Your own delivery database
- **Road Conditions:** Municipal road quality data

5. Continuous Improvement(future possibilities)

- **Online Learning:** Update model with new delivery data
- **Feedback Loop:** Delivery personnel can report road conditions
- **Reinforcement Learning Model:** that rewards on time deliveries and learns over time

What Makes Our Project Stand Out?

While giants like BlueDart, Delhivery, or Amazon focus on large-scale logistics, our solution optimizes hyper-local, multi-vehicle delivery routes—especially for smaller businesses or last-mile fleets in areas like Udupi.

Here's why we're different:

1. Real-World Road Networks, Not Just Straight-Line Distances

Others often use simplistic distance matrices

We integrate actual road networks (via OSMnx + OpenStreetMap) to model real travel times, one-way streets, and turn restrictions.

2. Time Windows + Dynamic Constraints

While big players optimize for speed or cost, we prioritize customer availability windows (e.g., "deliver between 3–5 PM")—key for local businesses (e.g., groceries, medicines) where timing matters.

3. ML-Powered Time Predictions

Current systems rely on fixed speed assumptions (e.g., "highway = 60 km/h", basically decides speed based on road type to calculate time in the time-cost matrix).

We're layering ML to predict delays using:

- Real-time traffic (via APIs like Google Maps)
- Weather (rain/floods impacting Udupi's roads)
- Historical patterns (e.g., school-hour congestion near Manipal)

This is a game-changer for reliability—especially where maps alone fail (e.g., rural roads, festivals).

4. Cost-Effective

Our solution is lightweight, customizable, and optimized for limited vehicles—no overkill enterprise software.

5. Transparency + Visualization

Drivers get turn-by-turn directions on real roads (not just GPS pins).
Drivers have the option to Re Optimize route mid delivery if required.

"Why Not Just Use Big Players?"

They're generic. Their algorithms ignore hyper-local nuances. They're expensive. Overkill for some businesses.

They don't adapt. Our ML layer learns from local data (Eg: Parkala roads are busy on Fridays)

We offer:

- I. Precision (real roads + time windows)
- II. Affordability (for small/medium fleets)
- III. Adaptability (ML learns local quirks)

"If you need a one-size-fits-all solution, use Amazon. If you want routes tailored to your customers, drivers, and city's roads—we're the answer."

Key Takeaways for Critics:

1. Highlight real-road accuracy (not theoretical distances).
2. Emphasize time windows (critical for customer satisfaction).
3. ML (where we surpass static systems).

big players like Amazon, FedEx, and Delhivery optimize deliveries. But:

- Their solutions are **not available** for smaller businesses
- They often ignore **local context (weather, regional roads)** in real-time
- Their models are **opaque and expensive**

We offer:

- **Transparent, affordable, modular tech**
- **Customizable ML layer** trained on specific regions
- A system that even a **small company** can adopt and run

Who are your customers?

- **Local logistics companies** in tier-2/3 cities
- **Government services** (postal or delivery-based schemes)
- **E-commerce startups** with growing delivery operations
- **NGOs** managing last-mile delivery in rural areas

Logistics firms must have complete insight into the supply chain in order to ensure accurate, on-time deliveries every time.

Send warnings and alerts when anything goes wrong with a shipment so that immediate action may be done if the shipment isn't arriving on time. Customers must be able to monitor their goods online and get reminders about when they will arrive, such as shipping notifications and ETAs.

Services Offered

The core is our **AI Route Optimization**, which includes multi-vehicle routing with time windows.

- Beyond that, we provide Smart Operations Management with live driver tracking using WebSocket technology ,
- instant delivery status updates ,
- and real-time weather updates.

1. For Tracking & Notifications, we have an intuitive Admin panel with Google Maps integration ,
2. a Driver mobile app with turn-by-turn navigation ,
3. a customer time slot selection system ,
4. and automated email notifications.

When **last-minute orders** come in, a driver becomes unavailable, or a significant traffic incident occurs, the administrator has real-time visibility. They can then trigger a **re-optimization** from the Admin Dashboard using our Google OR-Tools module, incorporating the new constraints or issues. This generates a revised optimized route that can be quickly assigned.

Route Optimization Models (Google OR-Tools) 🌟

Our project employs **Google OR-Tools** for route optimization, while we are exploring Deep Q Learning as a Reinforcement learning method.

Google OR-Tools is a powerful open-source software suite for solving combinatorial optimization problems, including Vehicle Routing Problems, or VRPs. We leverage its capabilities for both single and multi-vehicle optimization with time windows. This means we can efficiently plan routes for a fleet of vehicles while adhering to specific delivery time constraints for each parcel.



[The truly innovative aspect is our **Machine Learning Layer**. This layer modifies the travel time matrix used by OR-Tools based on various real-world factors. Instead of static estimations, our model learns from data to provide highly accurate travel times. This includes features like:

- Day of the week and month
- Whether it's a weekend, rush hour, or night time

- Environmental conditions like temperature, humidity, weather (rain, snow, etc.), wind speed, and visibility
- Even package characteristics like weight.

By incorporating these dynamic factors, our system moves beyond simplistic shortest-path algorithms to offer truly optimized, adaptive routes that account for real-world complexities. This results in significant improvements in on-time deliveries and minimization of travel time and distance.



Technologies Used & Future Improvements

- **Backend:** Node.js, Express.js, TypeScript, and Prisma ORM.
- **Frontend (Admin):** Next.js, React, TypeScript, and Tailwind CSS.
- **Frontend (Driver):** React Native and Expo.
- **Database:** PostgreSQL.
- **Google OR-Tools.**
- **APIs:** OpenStreetMap, Google Maps, and Weather API for real-time data.
- **Real-time Communication:** WebSocket for live location tracking.

Scalability of the solution for a large number of vehicles/deliveries.

"Our solution has been architected with scalability as a core principle.

- **Cloud-Native Database:** We've chosen **Neon DB for PostgreSQL** which is a serverless PostgreSQL database designed for scalability and efficiency. Its nature allows it to handle fluctuating loads and large datasets seamlessly. The fact that it's **already hosted** was a key reason for its selection, abstracting away much of the infrastructure management.
- **Modular API Design:** Our backend is separated into a **main API** and a dedicated **WebSockets API** for live location tracking. **This microservice-like approach means each component can scale independently based on demand.** The main API handles core business logic and database interactions, while the WebSockets API efficiently manages real-time, persistent connections for location updates.
- **Google OR-Tools:** This powerful library is inherently capable of solving large-scale Vehicle Routing Problems with many nodes and vehicles. While computation time increases with problem complexity, the library itself is optimized for performance.

- **Future-Proofing:** Our roadmap includes full cloud deployment with Docker containerization and CI/CD pipelines on platforms like AWS/GCP/Azure. This will enable **horizontal scaling** and elastic resource allocation to handle virtually any number of vehicles or deliveries."

Handling dynamic changes (e.g., last-minute orders, driver unavailability, traffic incidents).

- **Real-time Driver Tracking:** Our **WebSockets API** is crucial here. The driver app initiates **background location tracking** based on distance moved or time elapsed. This data is continuously sent to the WebSockets API, which then updates our main API and subsequently the database. Crucially, this real-time location is immediately reflected on the **Admin Portal's live tracking map**. This gives dispatchers instant awareness of driver positions and progress.
- **Admin Control & Re-optimization:** When last-minute orders come in, a driver becomes unavailable, or a significant traffic incident occurs, the administrator has real-time visibility. They can then trigger a **re-optimization** from the Admin Dashboard using our Google OR-Tools module, incorporating the new constraints or issues. This generates a revised optimized route that can be quickly assigned.
- **Real-time Weather Integration:** Our ML layer for OR-Tools incorporates live weather conditions, ensuring that initial route optimizations and subsequent re-optimizations consider factors like rain or high winds that impact travel times.
- **Instant Status Updates:** Drivers can immediately mark deliveries as completed or cancelled, updating the system and informing the admin portal and customers in real-time."

The difference and trade-offs between Deep Q Learning and Google OR-Tools.

Answer: "We are exploring two distinct models for optimization, each with unique strengths:

- **Google OR-Tools with ML Layer (Our Primary Model):** This is our production-ready core. OR-Tools is a robust, highly efficient library for solving complex Vehicle Routing Problems with precise constraints like time windows and capacities. Its strength lies in its proven ability to find optimal or near-optimal solutions quickly. The trade-off is that it requires explicit problem modeling.
- **Deep Q Learning (Research & Future Potential):** Our custom OpenAI Gym environment allows us to train an AI agent to learn optimal routing strategies through trial and error in a simulated environment. Its strength is its ability to learn complex, non-linear relationships and potentially discover novel optimization strategies without explicit programming of every rule. It's highly adaptive to dynamic and uncertain environments. The trade-offs include **significant training data requirements**, longer

training times, and the 'black-box' nature of deep learning, making interpretability challenging." "Our strategy is to leverage the proven power of OR-Tools for immediate, real-world application, while continuously researching and developing the Deep Q Learning model for future, potentially more adaptive and autonomous optimization capabilities."

Security measures for data and authentication.

- **JWT Tokens:** We use JSON Web Tokens for secure authentication between the client applications (admin dashboard, driver app) and our backend APIs. These tokens ensure that each request is authenticated and authorized, verifying the user's identity and permissions.
- **Bcrypt for Passwords:** All user passwords are securely hashed using bcrypt, an adaptive and robust hashing algorithm designed to resist brute-force attacks. This means we never store raw passwords, significantly enhancing user data security.
- **Role-Based Access Control:** Our Backend API enforces strict authorization policies, ensuring that administrators have access only to administrative functions, and drivers only to their assigned delivery information.
- **Database Security (PostgreSQL/Neon DB):** PostgreSQL itself has robust security features, and Neon DB provides additional layers of cloud-native database security, including isolation and secure connections."

USP (Unique Selling Proposition) of Our Project

1. **Intelligent, Adaptive Route Optimization with ML Layer:** While many platforms offer route optimization, our core differentiator is the **Machine Learning layer that dynamically modifies travel time estimates based on real-world, real-time factors.** We go beyond static map data to consider weather conditions, rush hour, day of the week, and even package weight. This results in routes that are not just mathematically optimal, but genuinely **adaptive and realistic**, leading to unparalleled accuracy in ETAs and significant fuel savings.
2. **Robust Real-time Operational Visibility:** Our system provides unparalleled real-time insights. The **dedicated WebSockets API** ensures seamless, low-latency live driver tracking, which is instantly reflected on the admin dashboard. This clear separation of real-time data handling from the main API ensures high performance and immediate operational awareness, crucial for dynamic decision-making.

3. **End-to-End Integrated Solution:** MargaDarshi isn't just a routing engine; it's a comprehensive ecosystem. From customer time slot selection and automated email notifications to a feature-rich admin dashboard for complete oversight and a user-friendly driver mobile app with background location tracking, we provide an integrated platform that addresses the entire delivery lifecycle.
4. **Scalability and Reliability through Modern Architecture:** By leveraging **Neon DB (a serverless PostgreSQL solution)** and adopting a modular API design, we've built a highly scalable and reliable system from the ground up. This ensures that MargaDarshi can efficiently grow with our clients' needs, from local businesses to large enterprises.